Hothyfa Elbeera
Senior Software Engineer | AI-Enabled Engineering | Cloud-Native Systems

Manchester, United Kingdom
Email: h.s.elbeera@gmail.com
Mobile: 07379095556
Website: www.hothyfa.com
LinkedIn | Github

Professional Summary

Senior Software Engineer with 7+ years of experience designing, developing and delivering cloud-native applications, distributed systems and AI-enabled software solutions across AWS and Azure.

Currently the sole Software Engineer responsible for the technical implementation of **Justice Redact at the UK Ministry of Justice**, an AI-assisted document redaction service operating within a highly regulated government environment. Responsible for engineering across application architecture, backend services, frontend development, NLP integration, asynchronous processing, automated testing and secure deployment.

Strong hands-on experience with Python, TypeScript, Node.js, React, FastAPI, cloud infrastructure and engineering automation. Experienced in building reusable software components, improving developer workflows, resolving complex engineering challenges and delivering secure, observable and maintainable systems.

Particularly interested in the practical application of AI to software engineering, including AI-assisted development, intelligent automation, engineering productivity and the evolution of cloud-native developer platforms.

Core Skills

Programming Languages: Python, TypeScript, JavaScript, SQL

AI & Intelligent Automation: Natural Language Processing (NLP), Microsoft Presidio, spaCy, custom entity recognition, AI-assisted document processing, AWS Comprehend integration, AI-assisted software development

Backend & APIs: FastAPI, Node.js, NestJS, Express.js, Fastify, REST APIs, GraphQL, distributed systems, asynchronous processing

Frontend Engineering: React, Next.js, Vue.js, Angular, React Native, HTML, CSS, Tailwind CSS

Cloud & Infrastructure: AWS, Azure, Kubernetes, Docker, AWS S3, SQS, RDS, ECR, CloudWatch, CI/CD, Github Actions

Databases & Storage: PostgreSQL, Microsoft SQL Server, MongoDB, DynamoDB, Redis, Firestore

Engineering Quality & Operations: Pytest, Vitest, Playwright, automated testing, TDD, observability, structured logging, OpenSearch, performance optimisation, security testing, Snyk

Architecture & Security: Cloud-native architecture, even-driven processing, API design, authentication, authorisation, data protection, secure-by-design engineering, reliability and fault tolerance

Engineering Practices: Agile, Scrum, Kanban, Git, Jira, technical documentation, cross-functional delivery, continuous improvement

Professional Experience

UK Ministry of Justice

Senior Full Stack Developer | February 2026 - Present
London (Remote)

Sole Software Engineer responsible for the application engineering and technical delivery of Justice Redact, an AI-assisted document processing and redaction service designed to help the Ministry of Justice staff identify, review and securely redact sensitive information from Subject Access Request documents.

Own end-to-end application development while working closely with Product, Data Science, DevOps, Architecture, Security and Design teams to deliver a scalable, maintainable and secure service.

Key Contributions & Achievements:

- **End-to-End Technical Ownership:** Lead hands-on engineering across a Next.js/React/TypeScript frontend, Python/FastAPI backend and reusable AI processing package, translating product requirements and architectural decisions into working software.
- **Scalable Distributed Processing:** Designed and implemented asynchronous document processing using AWS SQS, dedicated Kubernetes worker pods, PostgreSQL-backend job coordination, retries and dead-letter queues. Separated long-running processing from API request handling to support independent scaling and fault tolerance.
- **AI & NLP Integration:** Integrated AI-assisted entity detection using Microsoft Presidio, spaCy, custom recognisers and domain-specific post-processing. Worked alongside Data Science to incorporate NLP capabilities into an operational document review and redaction workflow.
- **Reusable Engineering Components:** Developed and integrated modular document processing capabilities for PDF extraction, entity recognition, document classification, review data generation and redaction application, supporting maintainability and future extension to additional document types.
- **Complex Workflow Engineering:** Implemented persistent document lifecycle management, versioned review decisions, asynchronous redaction operations and controlled export generation. Addressed concurrency, recovery and consistency challenges across frontend, backend and processing workers.
- **Performance & Scalability:** Implemented chunk-based processing and paginated review capabilities for large PDF documents, reducing the need to process and transfer entire document review datasets in a single operation. Investigated processing bottlenecks and benchmarked document extraction performance.
- **Engineering Automation:** Developed and maintained automated tests using Pytest, Vitest and Playwright, including regression coverage for workflow navigation, concurrent operations, failure handling and document export behaviour. Applied AI-assisted engineering during development and troubleshooting, validating changes through automated checks.

- **Cloud-Native Delivery:** Collaborated with DevOps on Docker-based services, Kubernetes deployments, AWS infrastructure, GitHub Actions pipelines and deployment configurations across development and staging environments.
- **Observability & Operational Reliability:** Implemented structured application and worker logging, operational event tracking and failure-handling mechanisms. Supported monitoring through OpenSearch and AWS infrastructure monitoring to improve troubleshooting and service reliability.
- **Security & Governance:** Implemented authenticated access, user-level document isolation and secure document handling. Worked with Architecture and Security stakeholders on access controls, data protection, vulnerability remediation and governance requirements within a sensitive government environment.
- **Cross-Functional Engineering:** Worked directly with product managers, interaction designers, data scientists, infrastructure engineers and technical architects to evaluate implementation approaches, communicate trade-offs and turn complex requirements into working solutions.

Core Technologies: Python, FastAPI, TypeScript, React, Next.js, Microsoft Presidio, spaCy, PyMuPDF, PostgreSQL, AWS S3, SQS, RDS, Docker, Kubernetes, GitHub Actions, OpenSearch, Pytest, Vitest, Playwright.

Claimsgate

Full Stack Developer | June 2025 - February 2026
London (Remote)

Responsible for designing, developing and delivering cloud-native applications at a growing digital claims management company. Contributed to application architecture, backend systems, infrastructure design and engineering improvements across the product lifecycle.

Key Contributions & Achievements:
- Designed and implemented scalable application architectures across AWS and Azure, with a focus on performance, reliability and infrastructure cost efficiency.
- Built and deployed containerised backend services using Node.js, Express.js and Docker, developing reusable APIs and service components.
- Developed cloud-native application capabilities using Firebase Functions, Firestore and Vue.js, supporting secure and responsive user experiences.
- Implemented API validation, secure data handling and encryption best practices to reduce application security risks.
- Developed monitoring and logging capabilities to improve application visibility, troubleshooting and operational support.
- Contributed to technical decision-making, system improvements and delivery coordination across product development.

Core Technologies: Node.js, Express.js, Vue.js, JavaScript, AWS, Azure, Docker, Firebase Functions, Firestore, REST APIs.

Shoory

Full Stack Developer | August 2024 - May 2025
Manchester (Remote)

Key engineer in the development of a multi-service platform covering ride-hailing, food delivery and parcel logistics. Worked alongside another Full Stack Developer to build web

applications, mobile applications, backend services and supporting infrastructure from an early stage.

Key Contributions & Achievements:
- Designed and developed web, mobile and backend applications using React, React Native, Node.js, Fastify and MongoDB.
- Built reusable API services supporting ride booking, delivery management, parcel logistics and real-time operational workflows.
- Implemented real-time tracking, secure authentication and authorisation mechanisms across platform services.
- Developed automated CI/CD pipelines to streamline application deployment and improve delivery consistency.
- Integrated push notification services and cross-platform application capabilities.
- Maintained technical documentation supporting system architecture, integration and ongoing development.

Core Technologies: React, React Native, Node.js, Fastify, MongoDB, REST APIs, CI/CD.

Union Realtime

Full Stack Developer | January 2022 - August 2024
London (Remote)

Developed and maintained scalable applications with a focus on backend engineering, API design, database performance, reliability and maintainability.

Key Contributions & Achievements:
- Developed full-stack applications using Angular, TypeScript, Python and Node.js.
- Designed and implemented modular REST APIs using NestJS, supporting maintainable and extensible backend services.
- Optimised SQL Server queries and database interactions to improve application responsiveness and processing efficiency.
- Implemented automated deployment processes and CI/CD pipelines to improve release consistency and reduce manual deployment effort.
- Produced technical documentation covering API behaviour, application architecture and engineering practices.

Core Technologies: Angular, TypeScript, Python, Node.js, NestJS, Microsoft SQL Server, T-SQL, REST APIs, CI/CD.

Education & Certification

- Certified AWS Cloud Practitioner
- Certified AWS AI Practitioner
- React & React Native Certification
- Querying Microsoft SQL Server with Transact-SQL
- Node.JS, Express.JS and Mongo DB Certification
- BA (Hons) in Business Management - Middlesex University

Professional Interests

AI-enabled software engineering, agentic development workflows, developer productivity, distributed systems, cloud-native platforms and secure, production-grade AI systems.